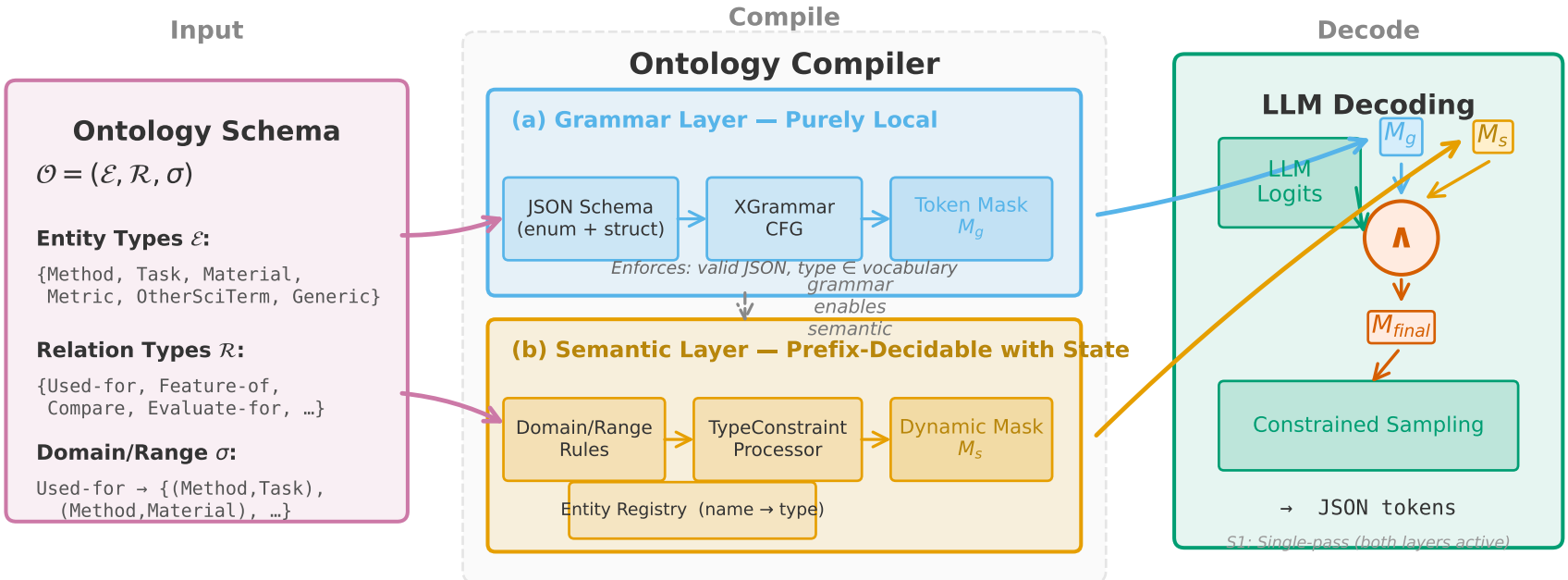

Input
Ontology Schema
$\mathcal{O} = (\mathcal{E}, \mathcal{R}, \sigma)$
Entity Types $\mathcal{E}$:
{Method, Task, Material,
Metric, OtherSciTerm, Generic}
Relation Types $\mathcal{R}$:
{Used-for, Feature-of,
Compare, Evaluate-for, …}
Domain/Range σ:
Used-for → {(Method,Task),
(Method,Material), …}
Compile
Ontology Compiler
(a) Grammar Layer — Purely Local
JSON Schema
(enum + struct)
XGrammar
CFG
Token Mask
M_g
Enforces: valid JSON, type ∈ vocabulary
grammar
enables
semantic
(b) Semantic Layer — Prefix-Decidable with State
Domain/Range
Rules
TypeConstraint
Processor
Dynamic Mask
M_s
Entity Registry (name → type)
Decode
LLM Decoding
LLM
Logits
M_g
M_s
∧
M_{final}
Constrained Sampling
→ JSON tokens
S1: Single-pass (both layers active)